Statistics

Lecture 23

Distribution of Sampling Statistics

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Text: Introduction to Probability and Statistics for Engineers and Scientists, Sheldon Ross (6th Ed)

Two parts of statistics

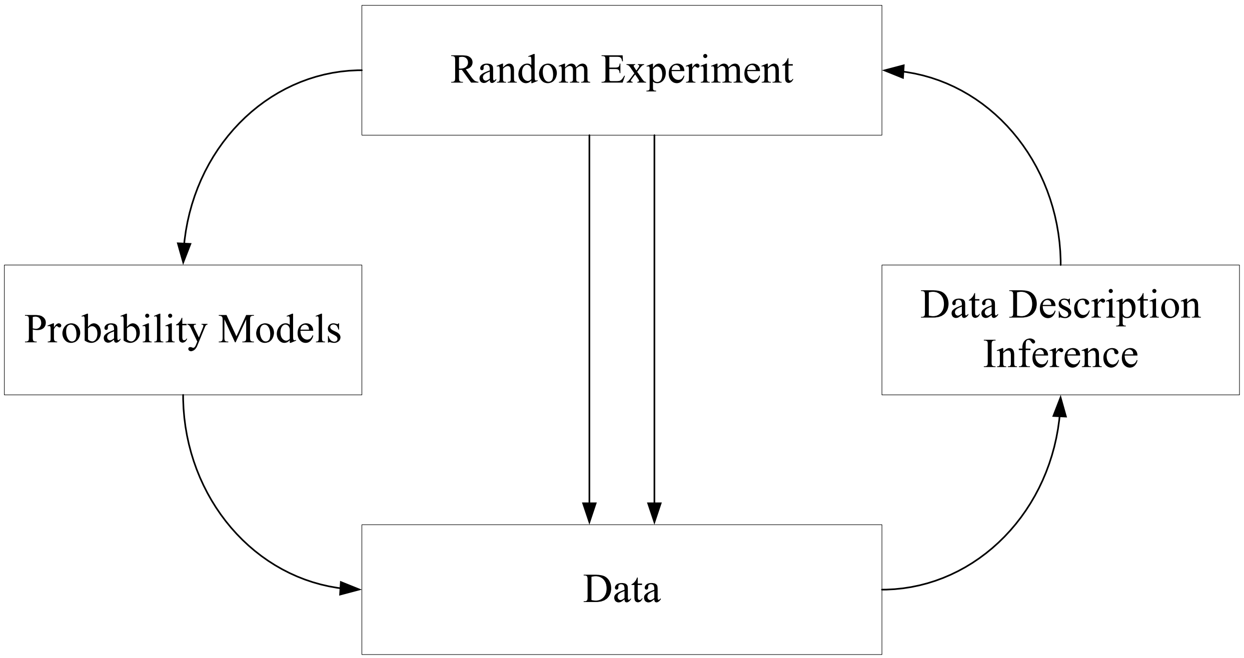
Descriptive Statistics

- **Description (presentation) of data**
  - Output: **tables** or **graphs or histograms**.
- **Summarization of data**
  - Output: numerical quantity from data (mean, median, variance, mode, etc.)

Inferential Statistics

- Involves techniques for
  - drawing conclusions
  - making inferences about a **population** from the **samples**.

Probability vs Statistics



Population and Sample

Population

- A **total** collection of elements being studied
- A group from which data (sample) is to be collected
- Complete set of individuals, objects, or scores of interest

Sample

- Population is often too large to examine.
- Sample is subset of a population.
  - Group of subjects selected from a population
- The sample must be informative about the total population (representative of that population).
  - Usually drawn in a totally **Random** fashion

TABLE 5.1
Quantitative SAT Scores and Final Examination Scores for 15 Introductory Psychology Students*

Student	*Quantitative SAT Score (X)*	*Final Examination Score (Y)*
1	595	68
2	520	55
3	715	65
4	405	42
5	680	64
6	490	45
7	565	56
8	580	59
9	615	56
10	435	42
11	440	38
12	515	50
13	380	37
14	510	42
15	565	53
Σ	8,010	772
	$\overline{X} = 534.00$	$\overline{Y} = 51.47$
	$s_X = 96.53$	$s_Y = 10.11$

**Note:* s_X and s_Y are sample standard deviations.

Measures of Central Tendency

Information about the **concentration** of scores in a distribution.

Statistics that describe the **center** of a set of data values:

- Mode
- Median
- Mean

Mean

Mean is the arithmetic average of the scores in distribution.

Symbol:

μ is for mean of population.

$$\mu = \frac{\sum x_i}{N}$$

N is size of the population.

$\bar{x}$ is for mean of sample.

$$\bar{x} = \frac{\sum x_i}{n}$$

n is size of the sample.

Sample Median

Definition

Order the values of a data set of size ***n*** from smallest to largest.

- If ***n*** is odd, the sample median is the value in position **(n + 1)/2**
- if ***n*** is even, it is the average of the values in positions **n/2** and **n/2 + 1**.

Median is actually **second quartile**.

{3, 6, 12, 18, 19, 21, 23} -> median = 4th datum = **18**.

{3, 6, 12, 18, 19, 21, 23, 25} -> median = (18 + 19) / 2 = **18.5**

Sample Mode

Mode is the most frequent score in a distribution.

Score	fr
783	**6**
785	4
786	2
788	2
789	2
790	2
791	3
792	2

← 783 is the most frequent score (6 times)

Mode of the data is 783

Example: Mode

A value i is said to be the mode of a discrete random variable X if it maximizes $P_X(x)$, probability mass function of X.

Find the mode of X and Y if

$$P_X(x) = \left(\frac{1}{2}\right)^x, x = 1, 2, \ldots$$

$$P_Y(y) = \left\langle \begin{matrix} 4 \\ y \end{matrix} \right\rangle \left(\frac{1}{4}\right)^y \left(\frac{3}{4}\right)^{4-y}, y = 0,1, \ldots, 4$$

Measures of variability (dispersion)

These statistics measure the **amount of scatter** in a data set.

Basic question: how widely scores are spread throughout the distribution ?

Measures of dispersion give us information about how much our variables vary from the mean.

- The measures of variation to be discussed:
  - mean deviation
  - variance
  - standard deviation

Mean Deviation

Deviation score is the difference between given score and the mean. $DS_i = (x_i - \bar{x})$

Mean deviation (MD) is the **average** of the **absolute values** of the **deviation scores**.

$$MD = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n} = \frac{\sum_{i=1}^{n} |DS_i|}{n}$$

Larger MD have greater variations !

Variance

- Using square instead of absolute.
- Variance is the **average** of the **sum of squared deviations** around the **mean**.

Symbol:

σ^2 is the variance of a population

$$\sigma^2 = \frac{SS}{N} = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

s^2 is the variance of a sample

$$s^2 = \frac{SS}{n-1} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Standard Deviation

Standard deviation is the **square root** of the **variance**.

Symbol:

σ is the standard deviation of a population

$$\sigma = \sqrt{\sigma^2}$$

s is the *sample standard deviation*

$$s = \sqrt{s^2}$$

Sample

Assumption about population and sample

- Underlying population distribution
- Measurable values in the population
  - Random variables with that distribution
  - Random variables are independent
- A sample is a collection of independent RVs
  - Having the same population distribution
  - RVs in a sample are IID

Sample

Definition. If $X_1, \ldots, X_n$ are independent random variables having a common distribution F, then we say that they constitute a *sample* (sometimes called a *random sample*) from the distribution F.

- F may not be completely specified
  - Use data to make inference about it
- F may be known but parameters are unknown
  - $F \sim N\ (\mu, \sigma^2)$ – both μ and σ^2 are unknown
- F may be completely unknown

Statistics

- **Statistic**: Random variable whose value is determined by the sample data

- **Distribution of sampling statistics**: Probability distribution of a statistic that arise from a sample

The Sample Mean

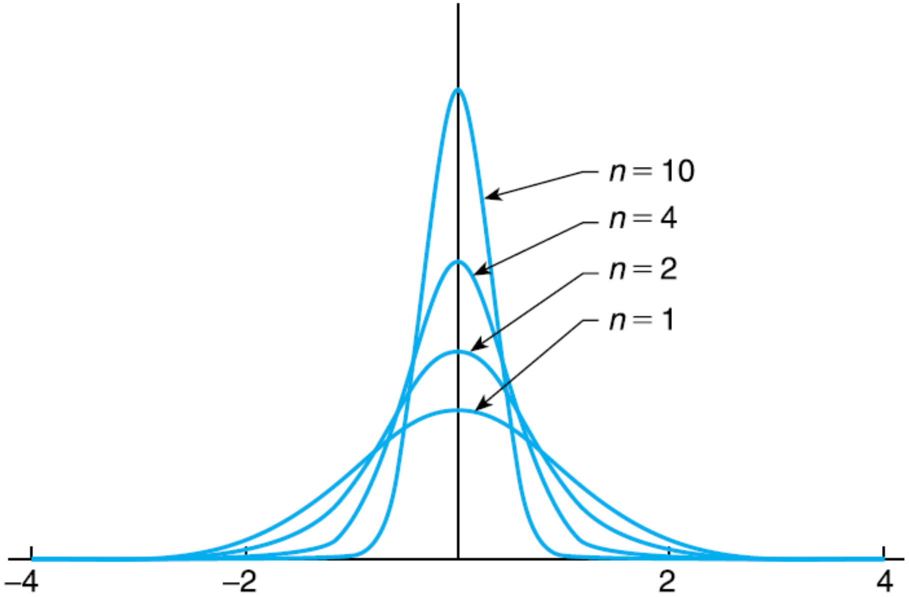
$$\overline{X} = \frac{X_1 + \cdots + X_n}{n}$$

expected value and variance are obtained as follows:

$$\begin{aligned} E[\overline{X}] &= E\left[\frac{X_1 + \cdots + X_n}{n}\right] \\ &= \frac{1}{n}(E[X_1] + \cdots + E[X_n]) \\ &= \mu \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\overline{X}) &= \text{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \cdots + \text{Var}(X_n)] \quad \text{by independence} \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$



The Central Limit theorem

Theorem 6.3.1 (The central limit theorem). Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables each having mean μ and variance σ^2. Then for n large, the distribution of

$$X_1 + \cdots + X_n$$

is approximately normal with mean $n\mu$ and variance $n\sigma^2$.

It follows from the central limit theorem that

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

is approximately a standard normal random variable; thus, for n large,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} < x\right\} \approx P\{Z < x\}$$

where Z is a standard normal random variable.

Example 6.3.a

Example 6.3.a. An insurance company has 25,000 automobile policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and standard deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

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$$P\{X > 8.3 \times 10^6\} = P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{8.3 \times 10^6 - 8 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$= P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{.3 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$\approx P\{Z > 3.51\}$$

$$\approx .00023$$

Example 6.3.b

Example 6.3.b. Civil engineers believe that W, the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1?

$$\begin{aligned} P_n &= P\{X_1 + \cdots + X_n \geq W\} \\ &= P\{X_1 + \cdots + X_n - W \geq 0\} \end{aligned}$$

$\sum_{i=1}^{n} X_i - W$ is approximately normal

$$E\left[\sum_{1}^{n} X_i - W\right] = 3n - 400$$

$$\mathrm{Var}\left(\sum_{1}^{n} X_i - W\right) = \mathrm{Var}\left(\sum_{1}^{n} X_i\right) + \mathrm{Var}(W) = .09n + 1{,}600$$

$$\begin{aligned} P_n &= P\left\{\frac{X_1 + \cdots + X_n - W - (3n - 400)}{\sqrt{.09n + 1{,}600}} \geq \frac{-(3n - 400)}{\sqrt{.09n + 1{,}600}}\right\} \\ &\approx P\left\{Z \geq \frac{400 - 3n}{\sqrt{.09n + 1{,}600}}\right\} \end{aligned}$$

Now $P\{Z \geq 1.28\} \approx .1$

$$\frac{400 - 3n}{\sqrt{.09n + 1{,}600}} \leq 1.28$$

or

$$n \geq 117$$

Approximation of Binomial RVs

$$X = X_1 + \cdots + X_n \qquad E[X_i] = p, \qquad \text{Var}(X_i) = p(1-p)$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

it follows from the central limit theorem that for n large

$$\frac{X - np}{\sqrt{np(1-p)}}$$

Example 6.3.c

Example 6.3.c. The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

parameters $n = 450$ and $p = .3$.

$P\{X = i\}$ as $P\{i - .5 < X < i + .5\}$ when applying the normal approximation

$$P\{X > 150.5\} = P\left\{\frac{X - (450)(.3)}{\sqrt{450(.3)(.7)}} \geq \frac{150.5 - (450)(.3)}{\sqrt{450(.3)(.7)}}\right\}$$

$$\approx P\{Z > 1.59\} = .06$$

Approximate Distribution of Sample Mean

Let $X_1, \ldots, X_n$ be a sample from a population having mean μ and variance σ^2.

$$\overline{X} = \sum_{i=1}^{n} X_i / n$$

$$\frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$ has approximately a standard normal distribution.

Example 6.3.d

Example 6.3.d. The weights of a population of workers have mean 167 and standard deviation 27.

(a) If a sample of 36 workers is chosen, approximate the probability that the sample mean of their weights lies between 163 and 170.
(b) Repeat part (a) when the sample is of size 144.

It follows from the central limit theorem that X is approximately normal with mean 167 and standard deviation $27/\sqrt{36} = 4.5$. Therefore, with Z being a standard normal random variable,

$$
\begin{aligned}
P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{4.5} < \frac{\overline{X}-167}{4.5} < \frac{170-167}{4.5}\right\} \\
&\approx P\{-.8889 < Z < .8889\} \\
&= P\{Z < .8889\} - P\{Z < -.8889\} \\
&= 2P\{Z < .8889\} - 1 \\
&\approx 2 * \ \mathrm{pnorm}(.8889) - 1 \\
&= 0.6259432
\end{aligned}
$$

For a sample of size 144, the sample mean will be approximately normal with mean 167 and standard deviation $27/\sqrt{144} = 2.25$. Therefore,

$$
\begin{aligned}
P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{2.25} < \frac{\overline{X}-167}{2.25} < \frac{170-167}{2.25}\right\} \\
&= P\left\{-1.7778 < \frac{\overline{X}-167}{4.5} < 1.7778\right\} \\
&\approx 2P\{Z < 1.7778\} - 1 \\
&= 2 * \text{ pnorm}(1.7778) - 1 \\
&= 0.9245633
\end{aligned}
$$

How large a sample is Needed

- Sample size for the validity of normal approximation
- If the underlying population distribution is normal
  - $\bar{X}$ is normal regardless of the sample size
- Rule of Thumb: for $n \geq 30$, normal approximation is valid
- Most cases, a much smaller sample size works
  - A sample size of 5 will often suffice

The Sample Variance

- $X_1, X_2, \ldots, X_n$ samples from a population with mean μ and variance σ^2, then
  - ❑ $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$ is called the sample variance
- $S^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\bar{X}^2\right)$
- $(n-1)E[S^2] = E[\sum_{i=1}^{n} X_i^2] - nE[\bar{X}^2]$

$$= nE[X_1^2] - nE[\bar{X}^2]$$

$$= nVar[X_1] + n(E[\bar{X}])^2 - nVar[\bar{X}] - n(E[\bar{X}])^2$$

$$= n\sigma^2 + n\mu^2 - \frac{n\sigma^2}{n} - n\mu^2 = (n-1)\sigma^2$$

$$E[S^2] = \sigma^2$$

➔ expectation of sample variance = population variance

Sampling Distribution from Normal Population

- $X_1, X_2, \dots, X_n$ random sample from normal population with Mean μ, and variance σ^2, then
  - $\bar{X} = \sum_{i=1}^{n} \frac{X_i}{n}$, and $S^2 = \sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{n-1}$
  - $E[\bar{X}] = \mu$, and $Var[\bar{X}] = \sigma^2/n$
  - Sum of independent normal RVs is normal, i.e.,

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

  - Therefore, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim Z$

Chi-Square Distribution

- If $Z_1, Z_2, \dots, Z_n$ are independent standard normal
- $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$
  - Chi-square distribution with n degrees of freedom
  - $X \sim \chi_n^2$
- If $X_1 \sim \chi_{n_1}^2$ and $X_2 \sim \chi_{n_2}^2$, then
- $(X_1 + X_2) \sim \chi_{n_1+n_2}^2$
- If $X \sim \chi_n^2$, and $\alpha(0,1)$, then the quantity $\chi_{\alpha,n}^2$ with n degree of freedom is defined as

$$P\left[X \geq \chi_{\alpha,n}^2\right] = \alpha$$

Joint Distribution of $\bar{X}$ and S^2

- $\bar{X}$ and S^2 are independent
- $(n-1)\frac{S^2}{\sigma^2}$ has a chi-square distribution with $(n-1)$ degrees of freedom
- Let $x_1, x_2, \dots, x_n$ is a random sample
  - $y_i = x_i - \mu, i = 1, 2, \dots, n$
  - $\bar{y} = \bar{x} - \mu$
- $\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2$
- $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i - \mu)^2 - n(\bar{x} - \mu)^2$
- $\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\sigma^2} = \sum_{i=1}^{n}\frac{(x_i-\bar{x})^2}{\sigma^2} + \frac{n(\bar{x}-\mu)^2}{\sigma^2}$

Joint Distribution of $\bar{X}$ and S^2

- $\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2 = \sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + \left(\frac{\sqrt{n}(\bar{x}-\mu)}{\sigma}\right)^2$
- Since $\frac{x_i-\mu}{\sigma}$ is standard normal
- $\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2$ has chi-square distribution with n degrees of freedom
- $\left(\frac{\sqrt{n}(\bar{x}-\mu)}{\sigma}\right)^2$ has a chi-square distribution with 1 degree of freedom
- $\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\sigma^2}$ has a chi-square distribution with $n-1$ degrees of freedom
- $\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\sigma^2} = (n-1)\frac{\sum_{i=1}^{n}\frac{(x_i-\bar{x})^2}{n-1}}{\sigma^2} = (n-1)\frac{S^2}{\sigma^2}$

the sum of two independent chi-square random variables is also chi-square with a degree of freedom equal to the sum of the two degrees of freedom.

Example 6.5.a

Example 6.5.a. The time it takes a central processing unit to process a certain type of job is normally distributed with mean 20 seconds and standard deviation 3 seconds. If a sample of 15 such jobs is observed, what is the probability that the sample variance will exceed 12?

$n = 15$ and $\sigma^2 = 9$

$$P\{S^2 > 12\} = P\left\{\frac{14S^2}{9} > \frac{14}{9} \cdot 12\right\}$$
$$= P\{\chi^2_{14} > 18.67\}$$
$$= 1 - .8221 \qquad \text{from Program 5.8.1a}$$
$$= .1779 \quad \blacksquare$$

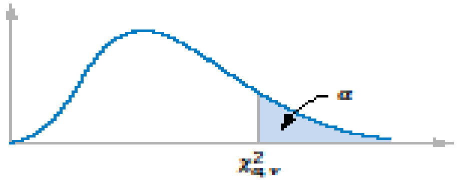

Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81	140.17

ν = degrees of freedom.

t-Distribution

- If $Z \sim N(0,1)$, and χ_n^2 are independent , then

$$T_n = \frac{Z}{\sqrt{\chi_n^2/n}}$$

is said to have a t-distribution with n degrees of freedom

- T_n has approximately the same distribution as Z
- $\frac{\sqrt{n}(\bar{X}-\mu)}{S} = \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{S/\sigma} = \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{\sqrt{S^2/\sigma^2}} = \frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{\sqrt{\frac{(n-1)S^2}{\sigma^2}/(n-1)}} = \frac{Z}{\sqrt{\chi_{n-1}^2/(n-1)}}$

Joint Distribution of $\bar{X}$ and S^2

- $X_1, X_2, \ldots, X_n$ is a sample from a normal population with mean μ
  - $\bar{X}$ is the sample mean
  - S is the sample standard deviation
- $\sqrt{n}(\bar{X} - \mu)/S$ has a t-distribution with $n - 1$ degrees of freedom
- $\frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t_{n-1}$

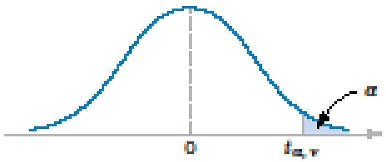

Table IV Percentage Points $t_{\alpha,\nu}$ of the t-Distribution

ν \ α	.40	.25	.10	.05	.025	.01	.005	.0025	.001	.0005
1	.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	.289	.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	.277	.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	.271	.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	.267	.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	.265	.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	.263	.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	.262	.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	.261	.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	.260	.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	.260	.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	.259	.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	.259	.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	.258	.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	.258	.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	.258	.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	.257	.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	.257	.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	.257	.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	.257	.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	.257	.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	.256	.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	.256	.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	.256	.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	.256	.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	.256	.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	.256	.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	.256	.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	.256	.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	.256	.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	.255	.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	.254	.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	.254	.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	.253	.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

ν = degrees of freedom.